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OVERVIEW OF ZIKA VIRUS

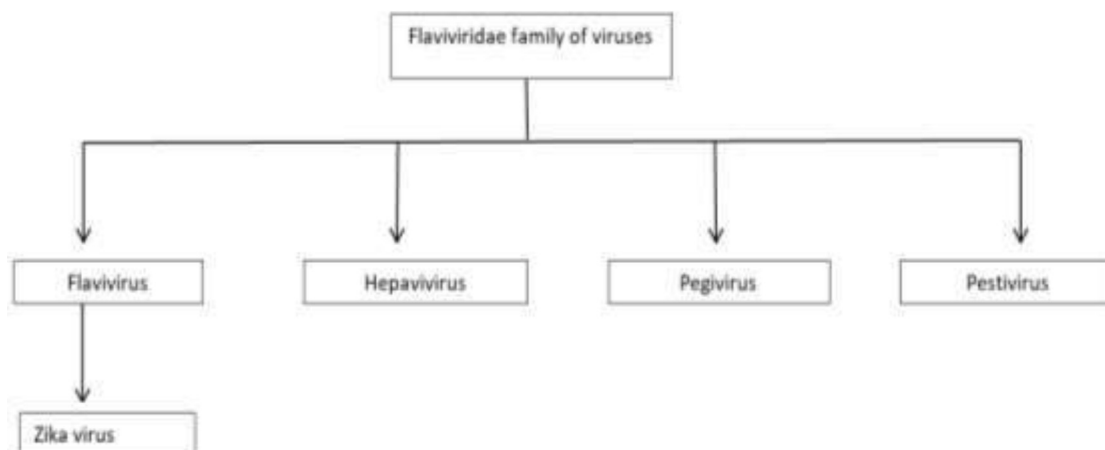
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The first cases of the zika virus were found in 1947 in Uganda and only minor outbreaks and few cases were reported in Asia and Africa during the periods ranging between 1960 to the early 21st century. An outbreak was diagnosed in the year 2007 on a place called Yap Island and similarly in 2013 epidemic was reported in French Polynesia. In Brazil, the zika virus was recognized in April 2015 and was reported to be the etiological agent of breakouts of an acute exanthematous illness that begun in the later period of 2014 in numerous cities of the Northeastern region of the country. After Brazil, the cases spread through South and Central America. In Continental US, the Maiden cases of Zika via local transmission were reported in the week of 24 July 2016. In the United States of America, on 19 April 2017 223 cases of suspected local transmission were reported.

Zika virus has emerged as a virus belonging to the family *Flaviviridae* that spreads *via* mosquito as the vector of the disease. The virus has an antigenic and phylogenetic identity that is connected to the Spondwenivirus. The diseases caused by the Zika virus are primarily arboviral. The Zika Virus structurally resembles an icosahedron. It is an enveloped RNA virus that is made up of a single strand. A covering of dense projections that constitute membrane and envelop glycoproteins surrounds the lipid envelope.





Etiology

The virus may spread through person-to-person contact, transfusion of blood, transplantation of organs, and perinatally and primarily are communicated via the bite of a female *Aedes, Aegypti* in addition to *Aedes albopictus* mosquitoes. Zika has symptoms that are connected to dengue, West Nile fever, and Yellow fever along with Japanese encephalitis viruses genetically.

Prognosis

Maximum cases of infection are mild and they normally get cured on their own but critical diseases of neurological have been reported that also comprise Guillain-Barre syndrome. There is also colossal concern that congenital disabilities like microcephaly, brain and eye abnormalities can be induced by the virus if it is acquired during the period of pregnancy. There have also been cases of acute myelitis and meningoencephalitis reported subsequently Zika virus infection. Hence a complete Neurologic examination is an advice when zika is suspected.

Physical effects

The maximum number of patients with acute Zika virus infections are either asymptomatic which ranges between 60-80% for they possess only mild symptoms estimated incubation phase between bites and symptoms is 2-14 days for mosquitoes bite. In asymptomatic infection, the most common symptoms include rashes, conjunctivitis, fever, and headache. Normally the rashes maculopapular with low-intensity fever and is a short period. Other frequent symptoms and signs include arthralgia, myalgia, and retro-orbital pain.



Treatment

The non-availability of a vaccine of a clinically tested drug to treat the infection coupled with the probability that another breakout might occur in the future defines an unachieved medical necessity. A large number of drug molecules are being under examination through the employment of repurposing study, screening through the compound library, and other new techniques during a short span of few years.

